

N

 LV.0 → LV.1 · AI ENGINEER

6 N-words that move you from **Level 0** → **1**.

Neural Networks, NLP, N-Grams & more. **Plain-English** definitions, one swipe each, plus the N-named tools, channels, and minds to follow.

↓ UP NEXT • TERMS

The N stack.

Neural Networks

NLP

NoSQL

N-Grams

Normalization

NanoLMs

N

Neural Networks

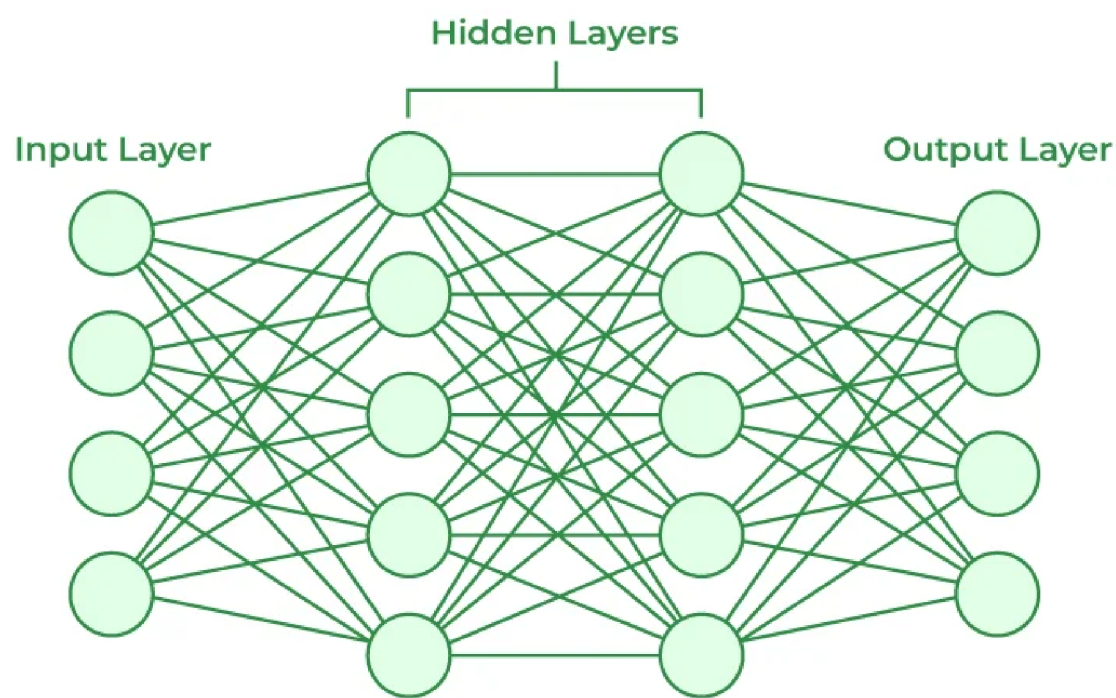
layers of neurons • patterns from data

Architectures inspired by the brain that stack layers of artificial neurons to **learn patterns directly from data**, the engine behind vision, speech, and modern generative AI.

• Computer vision

• NLP & LLMs

• Recommendations



INPUT → HIDDEN → OUTPUT

NATURAL LANGUAGE PROCESSING • LANGUAGE

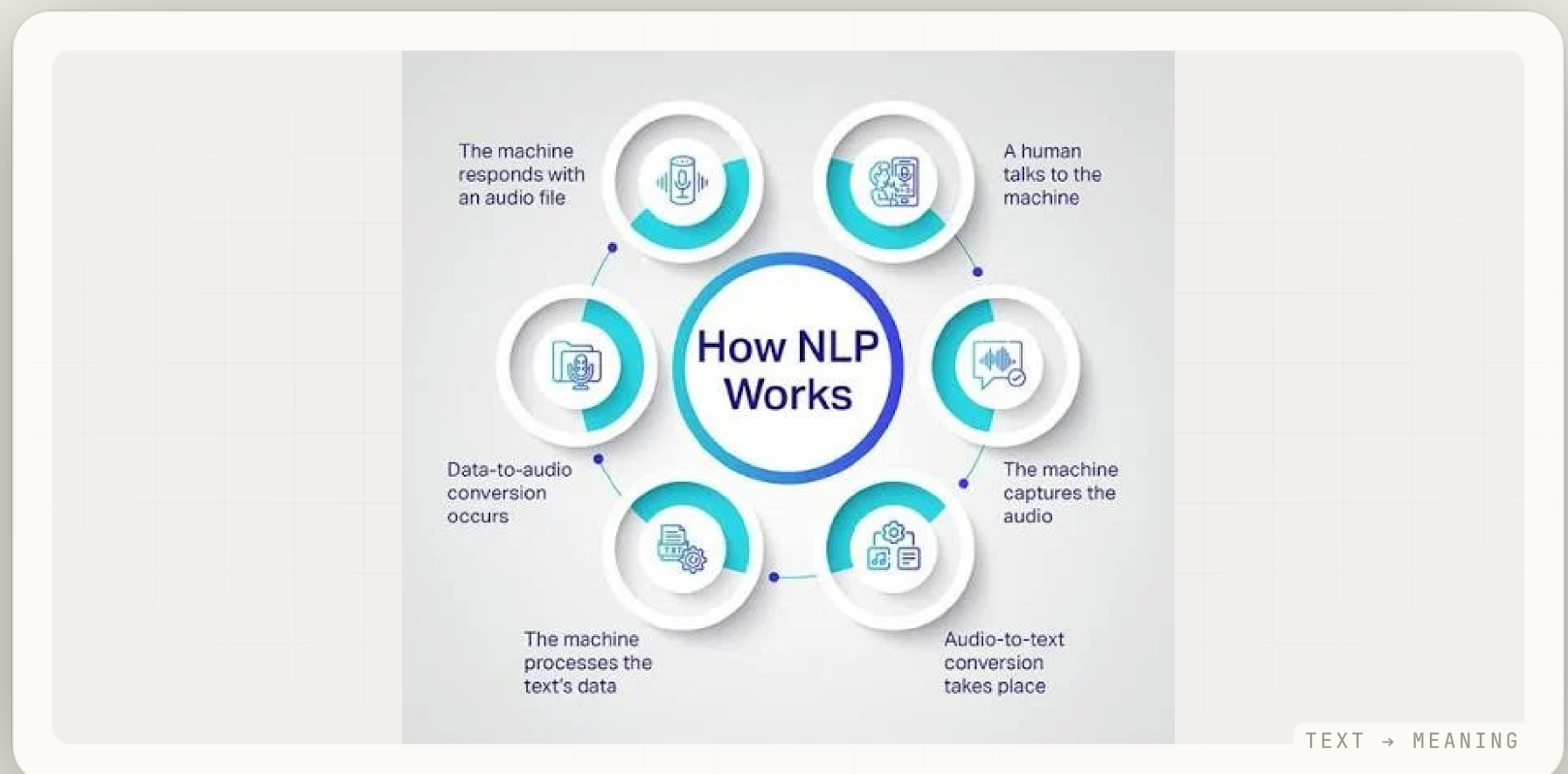
NLP

machines that read • write • understand

The field teaching machines to **understand, interpret, and generate human language**, powering chatbots, translation, search, sentiment analysis, and every modern LLM.

USED IN

- Chatbots
- Translation
- Search engines



N-GRAMS • STATISTICAL NLP

N-Grams

N tokens at a time • local context

Sequences of N consecutive words or tokens that **capture local context in text**, the backbone of classic language modeling, prediction, search, and spell correction.

USED IN

- Language modeling
- Spell correction
- Autocomplete

Text: "In consciousness studies the consciousness studies are fundamental"

Unigram	In	consciousness	studies	the	are	fundamental
Bigram	In consciousness	consciousness studies	studies the	the consciousness	studies are	are fundamental
Trigram	In consciousness studies	consciousness studies the	studies the consciousness	the consciousness studies	consciousness studies are	studies are fundamental

© AIML.com Research

UNIGRAM • BIGRAM • TRIGRAM

NORMALIZATION • DATA & TRAINING

Normalization

scale features • stabilize training

A preprocessing step that **scales features to a common range** before training, improving stability, speeding convergence, and stopping large-scale features from dominating learning.

USED IN

- Preprocessing
- Faster convergence
- Stable training

Scaling Formula:

$$X_{\text{scaled}} = \frac{X_i}{\|X\|}$$

Where:

- X_i is each individual value.
- $\|X\|$ represents the Euclidean norm (or length) of the vector X .
- Normalizes each sample to unit length.
- Useful for direction-based similarity metrics.

RAW → NORMALIZED RANGE

NanoLMs

useful language • tiny footprint

Compact language models built to **run on phones and edge hardware**, balancing real language ability with minimal memory, compute, and energy use.

USED IN

- On-device AI
- Edge hardware
- Mobile inference

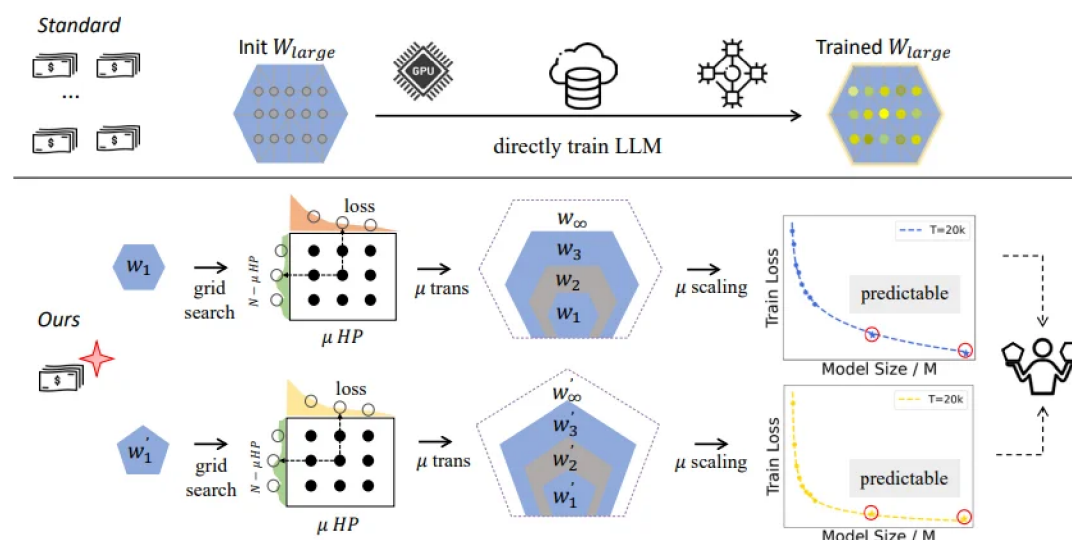


Figure 2. Illustration of standard LLM pre-training vs. nanoLM. **Top:** Directly pre-training LLM with high computational cost, large data, and distributed training. **Down:** loss prediction with μ Scaling. In this scenario, two different model designs, M and M' , are being compared. The process unfolds into four phases: 1) Shrink the width of the two models to base dimensions, denoted by w_1 and w'_1 , for grid search on μ Transferable HPs. 2) Choose a series of proxy models with small widths, then apply μ P for zero-shot HP Transfer for each model. 3) Train these proxy models, record the losses, and fit the scaling law. 4) Directly predict the loss at any given width without training large LLMs.

LLM VS NANOLM FOOTPRINT

NOSQL • INFRASTRUCTURE

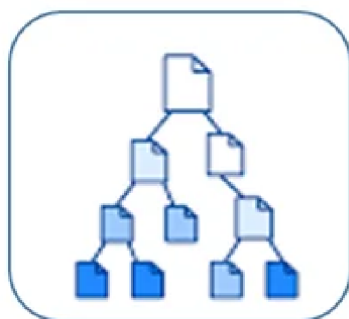
NoSQL

flexible schema • scale out

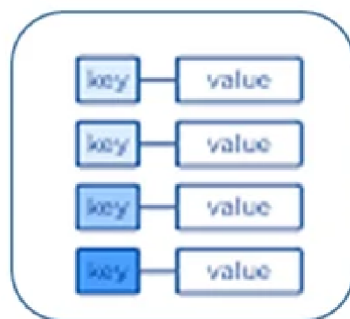
Non-relational databases that **store unstructured data at scale** with flexible schemas, widely used across AI, big data, and real-time analytics pipelines.

USED IN

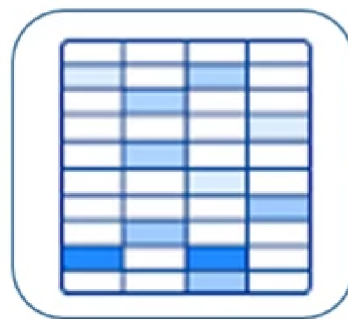
- Big data
- Real-time analytics
- AI pipelines



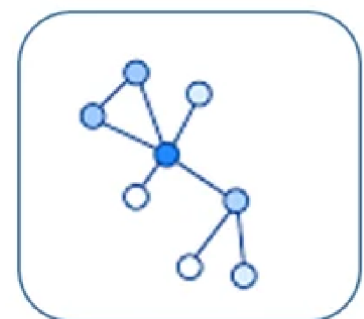
Document Store



Key-Value Store



Wide-Column Store



Graph Store

DOCUMENT • KEY-VALUE • GRAPH

↓ UP NEXT • TOOLS & COMPANIES

The N stack.

Four N-named tools and companies the whole AI stack quietly runs on, from **the GPUs that train it** to the databases and automations that ship it.

N

AI HARDWARE

Nvidia



GPUs • CUDA • the AI engine

The dominant GPU and AI-hardware company whose chips **power most large-scale AI training and inference** on Earth, from H100s to Blackwell.

1993

Founded by Jensen Huang

H100

Blackwell, the AI workhorse

CUDA

The moat every framework rides

WHAT IT IS

GPUs and the CUDA software stack for AI.

IN AI STACKS

Training clusters, inference, and edge.

WHY IT MATTERS

Trains and serves nearly all frontier models.

FIND THEM

[nvidia.com](https://www.nvidia.com)

AI RESEARCH NOTEBOOK

NotebookLM

 NotebookLM

Google • grounded in your docs

Google's research and note-taking tool that lets you upload sources and chat with **an AI grounded specifically in your documents**, including audio overviews.

Google

Built on Gemini models

Grounded

Answers only from your sources

Audio

Generates podcast-style overviews

WHAT IT IS

Source-grounded AI notebook and chat.

STANDOUT FEATURE

Auto audio overviews of your notes.

WHY DEVS USE IT

Reason over docs without hallucination.

FIND THEM

notebooklm.google.com

WORKFLOW AUTOMATION

n8n



open-source • no-code AI agents

An open-source automation platform that lets you **wire APIs, LLMs, and tools into agents** with little or no code, the glue layer for production AI workflows.

Open

Open-source, self-hostable core

No-code

Visual node-based builder

400+

Native app and API integrations

WHAT IT IS

Visual workflow and AI-agent builder.

IN AI STACKS

Connects LLMs to apps, data, and triggers.

WHY DEVS USE IT

Automate pipelines without glue scripts.

FIND THEM

n8n.io

GRAPH DATABASE

Neo4j

nodes • edges • GraphRAG



The leading graph database for **modeling and querying connected data**, increasingly used for knowledge graphs, GraphRAG, and relationship-aware retrieval.

2007

Pioneer of the graph database

Cypher

Its expressive query language

GraphRAG

Relationship-aware retrieval

WHAT IT IS

Native graph database for connected data.

IN AI STACKS

Knowledge graphs and GraphRAG pipelines.

WHY DEVS USE IT

Query relationships SQL can't reach.

FIND THEM

neo4j.com

↓ UP NEXT • CHANNELS TO FOLLOW

Learn from the best.

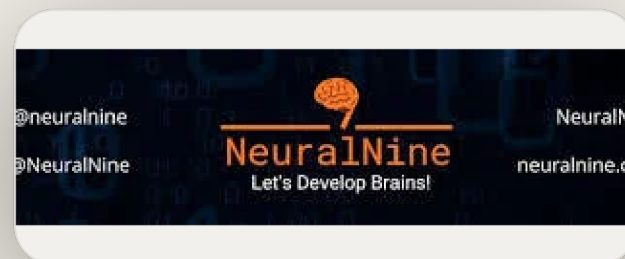
Two N-channels worth subscribing to, one drills
Python and ML by code-along, the other builds the
infrastructure intuition underneath it all.

N

YOUTUBE CHANNEL

NeuralNine

@neuralnine • Python + ML



Python-focused tutorials spanning machine learning, automation, and software dev, popular for **clear, code-along explanations** aimed at self-taught builders.

Python

ML, automation, and dev tutorials

Code-along

Build projects line by line

Free

Full tutorials on YouTube

BEST FOR

Hands-on Python and ML practice.

COVERS

ML, automation, and software dev.

TEACHING STYLE

Code-along, project-driven, no fluff.

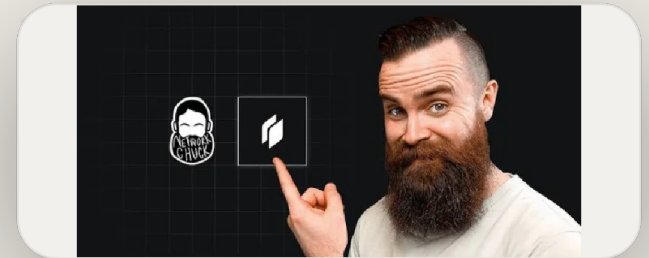
FIND THEM

youtube.com/@NeuralNine

YOUTUBE CHANNEL

NetworkChuck

@NetworkChuck • networking + IT



Networking, cybersecurity, and IT fundamentals taught with **hands-on, beginner-friendly energy**, a gateway before moving into AI infrastructure.

IT

Networking and cybersecurity

Hands-on

Labs, demos, and walkthroughs

Gateway

Foundations for AI infra topics

BEST FOR

Networking and infra fundamentals.

COVERS

Networking, security, Linux, cloud.

TEACHING STYLE

High-energy, hands-on, beginner-first.

FIND THEM

youtube.com/@NetworkChuck

↓ UP NEXT • PEOPLE TO FOLLOW

The N minds.

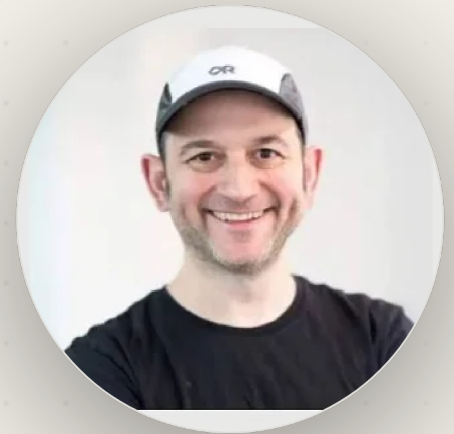
Five N-named researchers and educators **worth a follow**, from a Transformer co-author to the teachers who make it all click.

N

TRANSFORMER CO-AUTHOR

Noam Shazeer

Google DeepMind • @NoamShazeer



Co-author of "[Attention Is All You Need](#)," the paper that launched the Transformer. Co-founded Character.AI; now leads major LLM work including Gemini.

Transformer

Co-authored the founding paper

Gemini

Leads core LLM efforts at DeepMind

Character

Co-founded Character.AI

KNOWN FOR

The Transformer and modern LLM scaling.

BEST FOR

Foundational architecture thinking.

WHY FOLLOW

Signals where frontier models head next.

FIND THEM

linkedin.com/in/noam-shazeer-3b27288

AI RESEARCH SCIENTIST

Nikhil Naik

Meta Superintelligence • @nikhil_ai



Research scientist at Meta Superintelligence Labs working on LLMs, with an MIT PhD and published work across **computer vision, generative models, and NLP**.

Meta

Research at Superintelligence Labs

MIT

PhD, ex-Salesforce Research

Vision

CV, generative models, and NLP

KNOWN FOR

Research across vision and language.

BEST FOR

Research-grade ML and CV insight.

WHY FOLLOW

Frontier LLM work from inside Meta.

FIND THEM

[linkedin.com/in/nikdnaik](https://www.linkedin.com/in/nikdnaik)

GENERATIVE MODELS RESEARCHER

Nal Kalchbrenner

Google DeepMind • generative AI



Staff research scientist at Google DeepMind who **co-created WaveNet and PixelCNN**, foundational works in audio synthesis and generative modeling.

WaveNet

Co-created the audio model

PixelCNN

Pioneered autoregressive images

DeepMind

Staff research scientist

KNOWN FOR

WaveNet, PixelCNN, neural translation.

BEST FOR

Generative model fundamentals.

WHY FOLLOW

Deep roots in generative architectures.

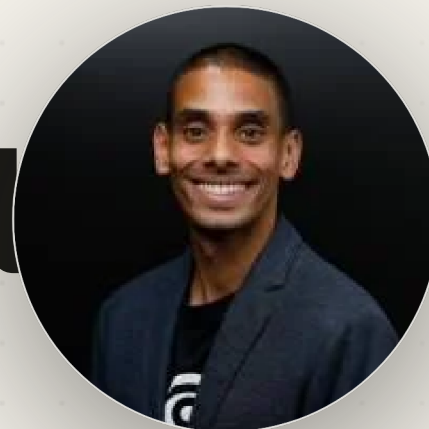
FIND THEM

[linkedin.com/in/kalchbrenner](https://www.linkedin.com/in/kalchbrenner)

AI EDUCATOR

Nicholas Renotte

YouTube • @nicholasrenotte



Data science specialist and YouTube educator teaching practical ML, deep learning, and generative AI in Python to hundreds of thousands of learners.

Python

Practical ML and DL tutorials

ex-IBM

Data science specialist

Hands-on

Buildable, end-to-end projects

KNOWN FOR

End-to-end ML and GenAI projects.

BEST FOR

Applied deep learning in Python.

WHY FOLLOW

Practical, build-it-yourself tutorials.

FIND THEM

linkedin.com/in/nicholasrenotte

AI AUTOMATION EDUCATOR

Nate Herk

@nateherk • AI Automation Society



Founder of AI Automation Society, teaching **n8n workflows and AI agents** to hundreds of thousands, and built an agency from \$0 to \$1M+ ARR.

n8n

Workflows and AI agents

\$1M+

Agency built from zero

No-code

Automation for non-engineers

KNOWN FOR

Teaching no-code AI automation.

BEST FOR

Shipping automations with n8n.

WHY FOLLOW

Practical agent and workflow builds.

FIND THEM

linkedin.com/in/nateherkelman

The N stack, in one card.

Save it. Drop a comment with the N-term you use most.

01 **Neural Networks** foundations

02 **NLP** language

03 **N-Grams** language

04 **Normalization** data

05 **NanoLMs** models

06 **NoSQL** infra

+ **Nvidia • NotebookLM** tools

+ **n8n • Neo4j** tools

► **NeuralNine • NetworkChuck** channel

@ **Noam • Nikhil • Nal • Nicholas • Nate** people



FOLLOW FOR THE FULL A TO Z
Prasanna Kulkarni
in/prasanna-kulkarni-032650180

0 Next